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TECHNICAL NOTE · INVERTERS

Deye three-phase hybrid inverter: how the BACKUP terminal works, and wiring a distant water pump onto backup

Models SUN-5/6/8/10/12K-SG04LP3 · example figures for the 10 kW model · Solar ESS Working Group · 2 Nov 2026

In short

- **The BACKUP (LOAD) terminal is live whenever the inverter is on**, with or without the grid.
- With the grid on, the inverter can supply solar/battery power to the BACKUP terminal and to grid-side loads at the same time, depending on the work mode.
- In a blackout an internal relay disconnects the grid side; the inverter supplies **only the BACKUP terminal** and never feeds PEA's lines.
- The BACKUP-terminal breaker for the 10 kW model is **32 A** per the manual, not 63 A.
- For a long cable protected only at its start, size it so the breaker **trips instantly** for a fault at the far end – not just for voltage drop.

1. Inside the inverter: one AC bus and a grid relay

- Inside the inverter, the solar and battery output, the GRID terminal and the BACKUP terminal share one AC bus. An internal relay connects that bus to the grid.
- **Grid on:** the relay is closed and grid power passes through to the BACKUP terminal, up to 45 A continuous.
- **Blackout:** the inverter senses it, opens the relay (anti-islanding) and supplies the BACKUP terminal from battery and solar, switching within milliseconds, like a UPS.
- **Grid back:** it synchronises, closes the relay and returns to pass-through automatically.
- **The BACKUP terminal is dead only when:** the inverter is off, the battery is at shutdown level with no sun, it has tripped on overload or a fault, or the BACKUP breaker is off.
- **Safety:** switching off the MDB main does not make BACKUP-side cables dead. Switch off the inverter, battery, DC isolators and BACKUP breaker, then lock out and tag out (LOTO).

2. Does it supply the grid side and the BACKUP terminal at the same time?

Work mode	With the grid on
Zero Export To Load	Supplies only the BACKUP-terminal loads (and charges the battery); not the MDB loads, and sells nothing
Zero Export To CT	Supplies the BACKUP loads and the MDB loads; a CT on the incoming main stops power flowing out; the grid tops up if needed
Solar Sell (on)	Surplus can be sold to the grid (used with either mode above)
Blackout (any mode)	Only the BACKUP terminal is supplied; the MDB loads go off

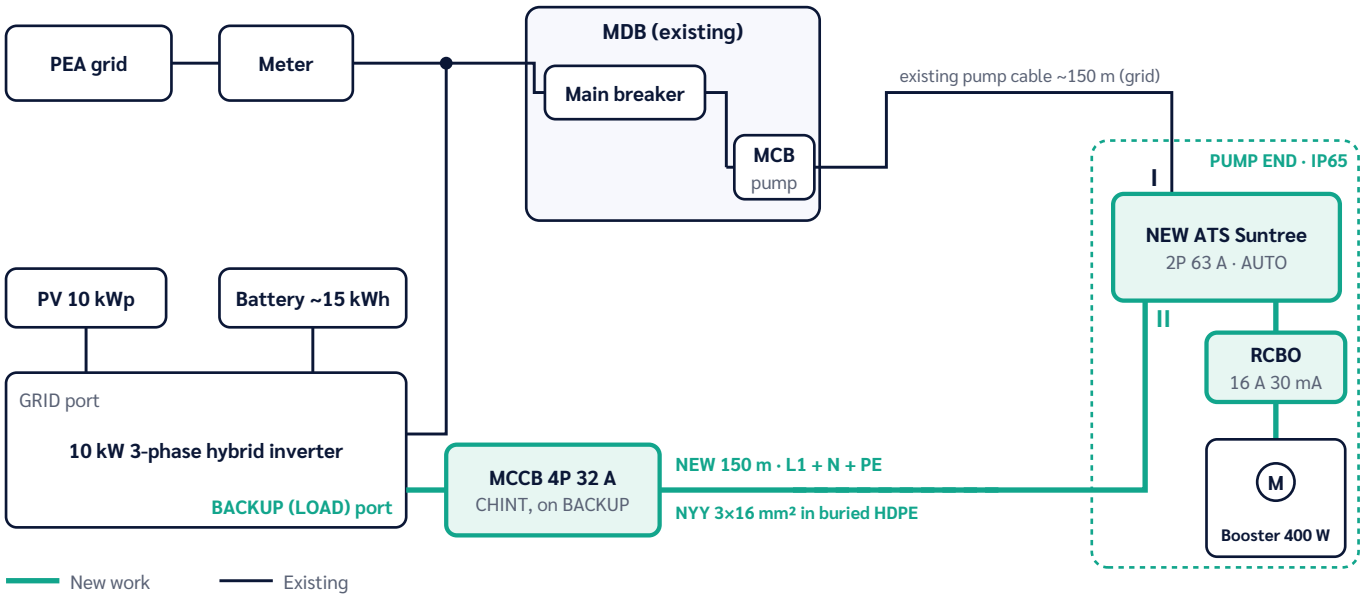
So where most loads are in the MDB yet the monitoring shows them running on solar and battery, the inverter is usually set to Zero Export To CT.

3. Figures to know (SUN-10K-SG04LP3)

Item	Value
Rated / maximum output	10 kW (15.2 A per phase) / 11 kW (22.7 A per phase)
Per phase	up to about 50% of rating (≈ 5 kW) on any one phase
Surge (off-grid)	2 × rated power for 10 s
Grid pass-through to BACKUP	45 A continuous
Recommended breakers	BACKUP terminal 32 A · GRID terminal 63 A
Max battery current	210 A

4. Example: a 400 W automatic booster pump 150 m from the inverter

The pump must keep running in a blackout. A 2P 63 A ATS and a 2P 16 A 30 mA RCBO sit next to the pump. Input I = the existing pump cable from the MDB (grid); input II = a new 150 m cable from the BACKUP terminal through an MCCB 4P 32 A (L1, N and PE used).



The new 150 m cable is protected only by the 32 A MCCB at its start, so a short circuit at the far end must exceed the MCCB instant-trip level (about 10 × 32 A = 320 A). For a 400 W pump (2.5 A running, 15 A starting) voltage drop is not the deciding factor.

NYY 3-core cable	Drop running / starting	Far-end fault	Result
3×6 mm²	1.2% / 7.2%	≈ 210 A	Fails: no instant trip
3×10 mm²	0.7% / 4.3%	≈ 330 A	Borderline
3×16 mm²	0.5% / 2.7%	≈ 460 A	Passes
3×16 mm² + 63 A MCCB	–	≈ 460 A < 630 A	Fails

- **MCCB 32 A, not 63 A:** the manual specifies 32 A, the terminal passes only 45 A, and 63 A would not trip fast for a far-end fault. Check the MCCB instant-trip setting is about 10 × In.
- **Buried cable:** NYY 0.6/1 kV in HDPE pipe ≥ 0.6 m deep; no THW in the ground. The PE core is also 16 mm², so an earth fault clears as fast as a short circuit.

- **ATS priority:** with the BACKUP cable on input I (priority) the pump always runs through the inverter and does not stop at all in a blackout (the inverter switches internally within milliseconds); the ATS moves to the direct grid cable only if the inverter is off or faulty. With the grid on input I, the pump stops for about 3–5 s while the ATS transfers, then restarts by itself.
- **No cross-feed:** the ATS is break-before-make and switches both L and N, so the BACKUP cable and the grid cable are never connected. Never join the BACKUP-terminal neutral to the MDB neutral bar.

5. Checklist before connecting loads to the BACKUP terminal

- Total backup load ≤ 10 kW and ≤ 5 kW per phase; no water heater or other large loads on the BACKUP terminal.
- Motors and pumps may start direct-on-line if the starting current is ≤ about twice the per-phase current (≈ 45 A); otherwise use a soft starter.
- BACKUP breaker 32 A (10–12 kW models); downstream cables and devices sized so faults clear instantly.
- The inverter bonds N–PE in off-grid mode, so RCDs/RCBOs work on battery power.
- Warning labels "Live from the battery during a grid outage" on every enclosure with BACKUP-side cables.

Sources

- Deye, Datasheet SUN-5/6/8/10/12K-SG04LP3: deyeinverter.com
- Deye, User Manual SUN-5/6/8/10/12K-SG04LP3 (recommended breakers, work modes): manualshive.com
- Voltage-drop and fault-current calculations: Solar ESS (copper 3.08 / 1.83 / 1.15 Ω/km for 6 / 10 / 16 mm², source impedance about 0.15 Ω).

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